For the use of a Registered Medical Practitioner only

PRESCRIBING INFORMATION

ESOZ Tablets

(Esomeprazole Magnesium Gastro Resistant 40 mg)

COMPOSITION

ESOZ 40 mg

Each gastro-resistant tablet contains: Esomeprazole Magnesium Amorphous equivalent to Esomeprazole…….………40 mg

Inactive components:

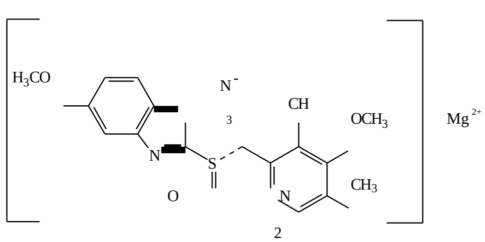
Sugar spheres , Hydroxyl propyl cellulose (HPC-L), Crospovidone PPXL-10, Povidone (PVP K-30), Macrogol 400, Purified talc, Isopropyl alcohol, Hypromellose phthalate(HP-55S), Hypromellose phthalate(HP-50), Diethylphalate, Acetone, Macrogol 6000, Methylene chloride, Microcrystalline cellulose PH101, Microcrystalline cellulose PH112, Sodium stearyl fumerate Macrogol 4000, and Opadry 03B86651 brown(HPMC 2910/Hypromellose 6cP,Tiatanium dioxide,Macrogol/PEG 400,Talc, Iron oxide red)

PRODUCT DESCRIPTION

ESOZ 40 mg: Light brick red to brown colored, oval, biconvex, film coated tablets with **'E6'** debossed on one side and plain on other side.

DESCRIPTION

ESOZ TABLETS contain esomeprazole magnesium. Chemically it is, (T-4)-Bis[5-methoxy-2-[(S)-[(4-methoxy-3,5-dimethyl-2-pyridinyl)methyl]sulfinyl]-1H-benzimidazolato]magnesium , a compound that inhibits gastric acid secretion. Esomeprazole is the S-isomer of omeprazole. Its molecular formula is $C_{34}H_{36}MgN_6O_6S_2$ with molecular weight of 713.12. The structural formula is:



Esomeprazole Magnesium

PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES

Pharmacodynamics

Esomeprazole is the S-isomer of omeprazole and reduces gastric acid secretion through a specific targeted mechanism of action. It is a specific inhibitor of the acid pump in the parietal cell. Both the R- and S-isomer of omeprazole have similar pharmacodynamic activity.

Site and mechanism of action

Esomeprazole is a weak base and is concentrated and converted to the active form in the highly acidic environment of the secretory canaliculi of the parietal cell, where it inhibits the enzyme H^+K^+-ATPase – the acid pump and inhibits both basal and stimulated acid secretion.

Effect on gastric acid secretion

After oral dosing with esomeprazole 20 mg and 40 mg the onset of effect occurs within one hour. After repeated administration with 20 mg esomeprazole once daily for five days, mean peak acid output after pentagastrin stimulation is decreased 90% when measured 6 - 7 hours after dosing on day five.

After five days of oral dosing with 20 mg and 40 mg of esomeprazole, intragastric pH above 4 was maintained for a mean time of 13 hours and 17 hours, respectively over 24 hours in symptomatic GERD patients. The proportion of patients maintaining an intragastric pH above 4 for at least 8, 12 and 16 hours respectively were for esomeprazole 20 mg 76%, 54% and 24%. Corresponding proportions for esomeprazole 40 mg were 97%, 92% and 56%.

Using AUC as a surrogate parameter for plasma concentration, a relationship between inhibition of acid secretion and exposure has been reported.

Therapeutic effects of acid inhibition

Healing of reflux oesophagitis with esomeprazole 40 mg occurs in approximately 78% of patients after four weeks, and in 93% after eight weeks.

One week treatment with esomeprazole 20 mg b.i.d. and appropriate antibiotics, results in successful eradication of *H. pylori* in approximately 90% of patients.

After eradication treatment for one week there is no need for subsequent monotherapy with antisecretory drugs for effective ulcer healing and symptom resolution in uncomplicated duodenal ulcers.

Other effects related to acid inhibition

During treatment with antisecretory medicinal products, serum gastrin increases in response to the decreased acid secretion. Also CgA increases due to decreased gastric acidity. The increased CgA level may interfere with investigations for neuroendocrine tumours.

Available published evidence suggests that proton pump inhibitors should be discontinued between 5 days and 2 weeks prior to CgA measurements. This is to allow CgA levels that might be spuriously elevated following PPI treatment to return to reference range.

An increased number of ECL cells possibly related to the increased serum gastrin levels, have been reported in some patients during long-term treatment with esomeprazole.

During long-term treatment with antisecretory drugs gastric glandular cysts have been reported to occur at a somewhat increased frequency. These changes are a physiological consequence of pronounced inhibition of acid secretion, are benign and appear to be reversible.

Decreased gastric acidity due to any means including proton pump inhibitors, increases gastric counts of bacteria normally present in the gastrointestinal tract. Treatment with proton pump inhibitors may lead to slightly increased risk of gastrointestinal infections such as *Salmonella* and *Campylobacter*.

Pharmacokinetics

Esomeprazole is acid labile and is administered orally as enteric-coated granules. *In vivo* conversion to the R-isomer is negligible. Absorption of esomeprazole is rapid, with peak plasma levels occurring approximately 1-2 hours after dose. The absolute bioavailability is 64% after a single dose of 40 mg and increases to 89% after repeated once-daily administration. For 20 mg esomeprazole the corresponding values are 50% and 68%, respectively. The apparent volume of distribution at steady state in healthy subjects is approximately 0.22 L/kg body weight. Esomeprazole is 97% plasma protein bound.

Food intake both delays and decreases the absorption of esomeprazole although this has no significant influence on the effect of esomeprazole on intragastric acidity.

Esomeprazole is completely metabolised by the cytochrome P450 system (CYP). The major part of the metabolism of esomeprazole is dependent on the polymorphic CYP2C19, responsible for the formation of the hydroxy- and desmethyl metabolites of esomeprazole. The remaining part is dependent on another specific isoform, CYP3A4, responsible for the formation of esomeprazole sulphone, the main metabolite in plasma.

The parameters below reflect mainly the pharmacokinetics in individuals with a functional CYP2C19 enzyme, extensive metabolisers.

Total plasma clearance is about 17 L/h after a single dose and about 9 L/h after repeated administration. The plasma elimination half-life is about 1.3 hours after repeated once-daily dosing. The pharmacokinetic properties of esomeprazole have been reported in doses up to 40 mg b.i.d. The area under the plasma concentration-time curve increases with repeated administration of esomeprazole. This increase is dose-dependent and results in a more than dose proportional increase in AUC after repeated administration. This time- and dose-dependency is due to a decrease of first pass metabolism and systemic clearance probably caused by an inhibition of the CYP2C19 enzyme by esomeprazole and/or its sulphone metabolite. Esomeprazole is completely eliminated from plasma between doses with no tendency for accumulation during once-daily administration.

The major metabolites of esomeprazole have no effect on gastric acid secretion. Almost 80% of an oral dose of esomeprazole is excreted as metabolites in the urine, the remainder in the faeces. Less than 1% of the parent drug is found in urine.

Special patient populations
Approximately 2.9 ±1.5% of the population lacks a functional CYP2C19 enzyme and are called as poor metabolisers. In these individuals the metabolism of esomeprazole is probably mainly catalysed by CYP3A4. After repeated once-daily administration of 40 mg esomeprazole, the mean area under the plasma concentration-time curve was approximately 100% higher in poor metabolisers than in subjects having a functional CYP2C19 enzyme (extensive metabolisers). Mean peak plasma concentrations were increased by about 60%. These findings have no implications for the posology of esomeprazole.

The metabolism of esomeprazole is not significantly changed in elderly subjects (71-80 years of age).

Following a single dose of 40 mg esomeprazole the mean area under the plasma concentration-time curve is approximately 30% higher in females than in males. No gender difference has been reported after repeated once-daily administration. These findings have no implications for the posology of esomeprazole.

Impaired organ function
The metabolism of esomeprazole in patients with mild to moderate liver dysfunction may be impaired. The metabolic rate is decreased in patients with severe liver dysfunction resulting in a doubling of the area under the plasma concentration-time curve of esomeprazole. Therefore, a maximum of 20 mg should not be exceeded in patients with severe dysfunction. Esomeprazole or its major metabolites do not show any tendency to accumulate with once-daily dosing.

No studies have been performed in patients with decreased renal function. Since the kidney is responsible for the excretion of the metabolites of esomeprazole but not for the elimination of the parent compound, the metabolism of esomeprazole is not expected to be changed in patients with impaired renal function.

Paediatric
Adolescents 12-18 years: Following repeated dose administration of 40 mg esomeprazole, the total exposure (AUC) and the time to reach maximum plasma drug concentration (t_{max}) in 12 to 18 year-olds was similar to that in adults for both esomeprazole doses.

INDICATIONS

ESOZ TABLETS are indicated for the following conditions: Gastroesophageal Reflux Disease

(GERD)

- treatment of erosive reflux esophagitis

Prolonged treatment after i.v. induced prevention of rebleeding of peptic ulcers. Treatment of

Zollinger Ellison Syndrome

ESOZ TABLETS are indicated in adolescents from the age of 12 years for: Gastroesophageal

Reflux Disease (GERD)

- treatment of erosive reflux esophagitis

DOSE AND METHOD OF ADMINISTRATION

ESOZ TABLETS are available in the strengths of 40 mg and may not be suitable for all dosage recommendations given below. Therefore, other suitable available strengths and/or dosage forms of esomeprazole should be used in such cases.

ESOZ TABLETS should be swallowed whole with liquid. The tablets should not be chewed or crushed. For the patients who have difficulty in swallowing the tablets can also be dispersed in half a glass of non carbonated water. No other liquid should be used as the enteric coating may be dissolved. Stir until the tablet disintegrate and drink the liquid with the pellets immediately or within 30 minutes. Rinse the glass with half a glass of water and drink. The pellets must not be chewed or crushed.

For patients who can not swallow, the tablet can be dispersed in non-carbonated water and administered through a gastric tube. It is important that the appropriateness of the selected syringe and tube is carefully tested.

Adults and Adolescents from the Age of 12 Years

Gastroesophageal Reflux Disease (GERD)

– *treatment of erosive reflux esophagitis*:
40 mg once daily for 4 weeks.
An additional 4 weeks treatment is recommended for patients in whom esophagitis has not healed or who have persistent symptoms.

Prolonged treatment after i.v. induced prevention of rebleeding of peptic ulcers
40 mg once daily for 4 weeks after i.v. induced prevention of rebleeding of peptic ulcers.

Treatment of Zollinger Ellison Syndrome
The recommended initial dosage is 40 mg twice daily. The dosage should then be individually adjusted and treatment continued as long as clinically indicated. Based on the reported clinical experience, the majority of patients can be controlled on doses between 80 to 160 mg esomeprazole daily. With doses above 80 mg daily, the dose should be divided and given twice daily.

Special Populations
Children below the age of 12 years
For the children younger than 12 years, any other approved formulation of esomeprazole available in market should be used.

Impaired Renal function
Dose adjustment is not required in patients with impaired renal function. Due to limited experience in patients with severe renal insufficiency, such patients should be treated with caution (see PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES).

Impaired Hepatic function
Dose adjustment is not required in patient with mild to moderate liver impairment. For patients with severe liver impairment, a dose of 20 mg should not be exceeded (see PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES).

Geriatric
Dose adjustment is not required in the elderly (see PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES).

Gender
No dose adjustment is necessary (see PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES).

CONTRAINDICATIONS

– Known hypersensitivity to esomeprazole, substituted benzimidazoles or to any other constituents of the formulation.
– Esomeprazole should not be used concomitantly with nelfinavir or atazanavir.

WARNINGS AND PRECAUTIONS

In the presence of any alarm symptom (e.g. significant unintentional weight loss, recurrent vomiting, dysphagia, haematemesis or melaena) and when gastric ulcer is suspected or present, malignancy should be excluded, as treatment with esomeprazole may alleviate symptoms and delay diagnosis.

Regular Surveillance
Patients on proton pump inhibitor treatment (particularly those treated for long term) should be kept under regular surveillance.

Subacute cutaneous lupus erythematosus (SCLE)
Proton pump inhibitors are associated with very infrequent cases of subacute cutaneous lupus erythematosus (SCLE). If lesions occur, especially in sun-exposed areas of the skin, and if accompanied by arthralgia, the patient should seek medical help promptly and the health care professional should consider stopping esomeprazole. SCLE after previous treatment with a proton pump inhibitor may increase the risk of SCLE with other proton pump inhibitors.

Patients on on-demand treatment should be instructed to contact their physician if their symptoms change in character. When prescribing esomeprazole for on-demand therapy, the implications for interactions with other pharmaceuticals, due to fluctuating plasma concentrations of esomeprazole should be considered.

When prescribing esomeprazole for eradication of *Helicobacter pylori,* possible drug interactions for all components in the triple therapy should be considered. Clarithromycin is a potent inhibitor of CYP3A4 and hence contraindications and interactions for clarithromycin should be considered when the triple therapy is used in patients concurrently taking other drugs metabolized via CYP3A4 such as cisapride.

Treatment with proton pump inhibitors may lead to slightly increased risk of gastrointestinal infections such as *Salmonella* and *Campylobacter*.

Co-administration of esomeprazole with atazanavir is not recommended. If the combination of atazanavir with a proton pump inhibitor is judged unavoidable, close clinical monitoring is recommended in combination with an increase in the dose of atazanavir to 400 mg with 100 mg of ritonavir; esomeprazole 20 mg should not be exceeded.

Atrophic gastritis has been noted occasionally in gastric corpus biopsies from patients treated long-term with omeprazole, of which esomeprazole is an enantiomer.

Fracture

Proton pump inhibitor, especially if used in high doses and over long durations (>1 year), may modestly increase the risk of hip, wrist, and spine fracture, predominantly in the elderly or in presence of other recognized risk factors. It has been reported that proton pump inhibitors may increase the overall risk of fracture by 10–40%. Some of this increase may be due to other risk factors. Patients at risk of osteoporosis- should receive care according to current clinical guidelines and they should have an adequate intake of vitamin D and calcium.

Hypomagnesaemia

Severe hypomagnesemia has been reported in patients treated with PPI like esomeprazole for at least three months, and in most cases for a year. Serious manifestations of hypomagnesemia such as fatigue, tetany, delirium, convulsions, dizziness and ventricular arrhythmia can occur but they may begin insidiously and be overlooked. In most affected patients hypomagnesemia improved after magnesium replacement and discontinuation of the PPI.

For patients expected to be on prolonged treatment or who take PPIs with digoxin or drugs that may cause hypomagnesemia (e.g., diuretics), health care professionals should consider monitoring magnesium levels before starting PPI treatment and periodically during treatment (see SIDE EFFECTS/ADVERSE REACTIONS).

Drugs which induce CYP2C19 or CYP3A4 (such as St John's Wort or rifampin) can substantially decrease esomeprazole concentrations. Avoid concomitant use of esomeprazole with St John's Wort, or rifampin. Esomeprazole is a CYP2C19 inhibitor. When starting or ending treatment with esomeprazole, the potential for interactions with drugs metabolised through CYP2C19 should be considered. An interaction is observed between clopidogrel and omeprazole. As a precaution, concomitant use of esomeprazole and clopidogrel should be discouraged.

Interference with laboratory tests

Increased chromogranin A (CgA) level may interfere with investigations for neuroendocrine tumors. If the patient(s) are due to have a test on Chromogranin A level, Esomeprazole treatment should be stopped for at least 5 days before CgA measurements to avoid this interference (see Pharmacodynamics). If CgA and gastrin levels have not returned to reference range after initial measurement, measurements should be repeated 14 days after cessation of PPI treatment.

Literature suggests that concomitant use of PPIs with methotrexate (primarily at high dose) may elevate and prolong serum levels of methotrexate and/or its metabolite, possibly leading to methotrexate toxicities. In high-dose methotrexate administration a temporary withdrawal of the PPI may be considered in some patients.

Vitamin B12 deficiency

Daily treatment with any acid-suppressing medications over a long period of time (e.g., longer than 3 years) may lead to malabsorption of cyanocobalamin (vitamin B12) caused by hypo- or a-chlorhydria. Rare reports of of cyanocobalamin deficiency occurring with acid-suppressing therapy have been reported in literature. This diagnosis should be considered if clinical symptoms consistent with cyanocobalamin deficiency are observed.

Clostridium difficile associated diarrhea

Published observational studies suggest that PPI therapy may be associated with an increased risk of *Clostridium difficile* associated diarrhea, especially in hospitalized patients. This diagnosis should be considered for diarrhea that does not improve (see SIDE EFFECTS/ADVERSE REACTIONS). Patients should use the lowest dose and shortest duration of PPI therapy appropriate to the condition being treated.

Clostridium diffficile associated diarrhea (CDAD) has been reported with use of nearly all antibacterial agents. For more information specific to antibacterial agents (clarithromycin and amoxicillin) indicated for use in combination with esomeprazole, refer to WARNINGS and PRECAUTIONS sections of those package inserts.

Effects on ability to drive and use machines: Esomeprazole has minor influence on the ability to drive and use machines. Adverse reactions such as dizziness (uncommon) and blurred vision (rare) has been reported (see SIDE EFFECTS/ADVERSE REACTIONS). If affected patients should not drive or use machines.

This product contains sucrose as an excipient. Patients with rare hereditary problems of fructose intolerance, glucose-galactose malabsorption or sucrase-isomaltase insufficiency should not take this product.

DRUG INTERACTIONS

Effects of esomeprazole on the pharmacokinetics of other drugs

Medicinal products with pH dependent absorption

The decreased intragastric acidity during treatment with esomeprazole, might increase or decrease the absorption of drugs if the mechanism of absorption is influenced by gastric acidity. In common with the use of other inhibitors of acid secretion or antacids, the absorption of ketoconazole, itraconazole and erlotinib can decrease and the absorption of digoxin can increase during treatment with esomeprazole. Concomitant treatment with omeprazole (20 mg daily) and digoxin in healthy subjects increased the bioavailability of digoxin by 10%. Digoxin toxicity has been rarely reported. However, caution should be exercised when esomeprazole is given at high doses in elderly patients. Therapeutic drug monitoring of digoxin should then be reinforced.

Esomeprazole may interfere with absorption of iron salts where gastric pH is an important determinant of bioavailability.

Omeprazole has been reported to interact with some protease inhibitors. The clinical importance and the mechanisms behind these reported interactions are not always known. Increased gastric pH during omeprazole treatment may change the absorption of the protease inhibitors. Other possible interaction mechanisms are via inhibition of CYP2C19.

For atazanavir and nelfinavir, decreased serum levels have been reported when given together with omeprazole and concomitant administration is not recommended. It has been reported that co-administration of omeprazole (40 mg once daily) with atazanavir 300 mg/ritonavir 100 mg resulted in a substantial reduction in atazanavir exposure (approximately 75% decrease in AUC, C_{max} and C_{min}). Increasing the atazanavir dose to 400 mg did not compensate for the impact of omeprazole on atazanavir exposure. The co-administration of omeprazole (20 mg qd) with atazanavir 400 mg/ritonavir 100 mg resulted in a decrease of approximately 30% in the atazanavir exposure as compared with the exposure observed with atazanavir 300 mg/ritonavir 100 mg qd without omeprazole 20 mg qd. Co-administration of omeprazole (40 mg qd) reduced mean nelfinavir AUC, C_{max} and C_{min} by 36–39% and mean AUC, C_{max} and C_{min} for the pharmacologically active metabolite M8 was reduced by 75-92%. For saquinavir (with concomitant ritonavir), increased serum levels (80-100%) have been reported with concomitant use of omeprazole (40 mg qd).

Treatment with omeprazole 20 mg qd had no effect on the exposure of darunavir (with concomitant ritonavir) and amprenavir (with concomitant ritonavir). Treatment with esomeprazole 20 mg qd had no effect on the exposure of amprenavir (with and without concomitant ritonavir). Treatment with omeprazole 40 mg qd had no effect on the exposure of lopinavir (with concomitant ritonavir). Due to the similar pharmacodynamic effects and pharmacokinetic properties of omeprazole and esomeprazole, concomitant administration with esomeprazole and atazanavir is not recommended and concomitant administration with esomeprazole and nelfinavir is contraindicated.

Drugs metabolised by CYP2C19
Esomeprazole inhibits CYP2C19, the major esomeprazole metabolising enzyme. Thus, when esomeprazole is combined with drugs metabolised by CYP2C19, such as diazepam, citalopram, imipramine, clomipramine, phenytoin etc., the plasma concentrations of these drugs may be increased and a dose reduction could be needed. This should be considered especially when prescribing esomeprazole for on-demand therapy. A 45% decrease in clearance of the CYP2C19 substrate diazepam has been reported with concomitant use of 30 mg esomeprazole. A 13% increase in trough plasma levels of phenytoin has been reported with concomitant use of 40 mg esomeprazole in epileptic patients. It is recommended to monitor the plasma concentrations of phenytoin when treatment with esomeprazole is introduced or withdrawn. Omeprazole (40 mg once daily) increased voriconazole (a CYP2C19 substrate) C_{max} and AUC by 15% and 41%, respectively.

It has been reported that coagulation times were within the accepted range on concomitant use of 40 mg esomeprazole in patients receiving warfarin. However, a few isolated cases of elevated INR of clinical significance have been reported during concomitant treatment. Increases in INR and prothrombin time may lead to abnormal bleeding and even death. Monitoring is recommended when initiating and ending concomitant esomeprazole treatment during treatment with warfarin or other coumarine derivatives.

A 32% increase in area under the plasma concentration-time curve (AUC) and a 31% prolongation of elimination half-life ($t_{1/2}$) but no significant increase in peak plasma levels of cisapride has been reported with concomitant use of 40 mg esomeprazole. The slightly prolonged QTc interval has been reported in patient receiving cisapride alone and was not further prolonged in combination with esomeprazole.

No clinically relevant effects on the pharmacokinetics of amoxicillin or quinidine have been reported with the use of esomeprazole.

Concomitant administration of esomeprazole and either naproxen or rofecoxib did not identify any clinically relevant pharmacokinetic interactions.

Omeprazole acts as an inhibitor of CYP 2C19. Increased C_{max} and AUC of cilostazol by 18% and 26%, respectively have been reported with the use of omeprazole (40 mg daily for one week). C_{max} and AUC of one of its active metabolites, 3, 4-dihydrocilostazol (which has 4-7 times the activity of cilostazol) have been reported to increase by 29% and 69% respectively. Co-administration of cilostazol with esomeprazole is expected to increase concentrations of cilostazol and its above mentioned active metabolite. Therefore a dose reduction of cilostazol from 100 mg b.i.d. to 50 mg b.i.d. should be considered.

A pharmacokinetic (PK)/ pharmacodynamic (PD) interaction between clopidogrel (300 mg loading dose/75 mg daily maintenance dose) and esomeprazole (40 mg p.o.daily) has been reported to decrease exposure to the active metabolite of clopidogrel and thus decreased maximum inhibition of (ADP induced) platelet aggression.

A decreased exposure of the active metabolite of clopidogrel has been reported with co-administration of clopidogrel and fixed dose combination of esomeprazole 20 mg + ASA 81 mg versus clopidogrel alone. However, the maximum levels of inhibition of (ADP induced) platelet aggregation was reported to be the same in the clopidogrel and the clopidogrel + the combined (esomeprazole + ASA) product groups.

The exposure to the active metabolite of clopidogrel has been reported to decrease by 46% (Day 1) and 42% (Day 5) when clopidogrel (300 mg loading dose followed by 75 mg/day) and omeprazole (80 mg at the same time as clopidogrel) were administered together for 5 days. Mean inhibition of platelet aggregation (IPA) was diminished by 47% (24 hours) and 30% (Day 5) when clopidogrel and omeprazole were administered together. It has also been reported that administration of clopidogrel and omeprazole at different times did not prevent their interaction, which is likely to be driven by the inhibitory effect of omeprazole on CYP2C19. Inconsistent information on the clinical implications of this PK/PD interaction in terms of major cardiovascular events has been reported.

Unknown mechanism

When given together with PPIs, methotrexate levels have been reported to increase in some patients. In high-dose methotrexate administration a temporary withdrawal of esomeprazole may need to be considered.

Concomitant administration of esomeprazole and tacrolimus may increase the serum levels of tacrolimus.

Effects of other drugs on the pharmacokinetics of esomeprazole

Esomeprazole is metabolised by CYP2C19 and CYP3A4. A doubling of the exposure (AUC) to esomeprazole has been reported with concomitant use of esomeprazole and a CYP3A4 inhibitor, clarithromycin (500 mg b.i.d.). Concomitant administration of esomeprazole and a combined inhibitor of CYP2C19 and CYP3A4 may result in more than doubling of the esomeprazole exposure. The CYP2C19 and CYP3A4 inhibitor voriconazole increased omeprazole AUC by 280%. A dose adjustment of esomeprazole is not regularly required in either of these situations. However, dose adjustment should be considered in patients with severe hepatic impairment and if long-term treatment is indicated.

Drugs known to induce CYP2C19 or CYP3A4 or both (such as rifampicin and St. John's wort) may lead to decreased esomeprazole serum levels by increasing the esomeprazole metabolism.

Combination Therapy with Clarithromycin

Co-administration of esomeprazole, clarithromycin, and amoxicillin has resulted in increases in the plasma levels of esomeprazole and 14-hydroxyclarithromycin.

Interactions with Investigations of Neuroendocrine Tumors

Drug-induced decrease in gastric acidity results in enterochromaffin-like cell hyperplasia and increased Chromogranin A levels which may interfere with investigations for neuroendocrine tumors.

SIDE EFFECTS/ADVERSE REACTIONS

The following adverse drug reactions have been identified or suspected for esomeprazole. None was found to be dose-related.

Blood and lymphatic system disorders *Rare*: Leukopaenia, thrombocytopaenia *Very rare*: Agranulocytosis, pancytopaenia

Immune system disorders

Rare: Hypersensitivity reactions e.g. fever, angioedema and anaphylactic reaction/shock

Metabolism and nutrition disorders

Uncommon: Peripheral oedema
Rare: Hyponatraemia
Not known: Hypomagnesaemia; severe hypomagnesaemia can correlate with hypocalcaemia. Hypomagnesaemia may also be associated with hypokalaemia.

Psychiatric disorders

Uncommon: Insomnia
Rare: Agitation, confusion, depression
Very rare: Aggression, hallucinations

Nervous system disorders

Common: Headache
Uncommon: Dizziness, paraesthesia, somnolence

Rare: Taste disturbance

Eye disorders
Rare: Blurred vision

Ear and labyrinth disorders
Uncommon: Vertigo

Respiratory, thoracic and mediastinal disorders
Rare: Bronchospasm

Gastrointestinal disorders
Common: Abdominal pain, constipation, diarrhoea, flatulence, nausea/vomiting, fundic gland polyps (benign)
Uncommon: Dry mouth
Rare: Stomatitis, gastrointestinal candidiasis
Not known: Microscopic colitis

Hepatobiliary disorders
Uncommon: Increased liver enzymes
Rare: Hepatitis with or without jaundice
Very rare: Hepatic failure, encephalopathy in patients with pre-existing liver disease

Skin and subcutaneous tissue disorders *Uncommon*: Dermatitis, pruritus, rash, urticaria *Rare*: Alopecia, photosensitivity
Very rare: Erythema multiforme, Stevens-Johnson syndrome, toxic epidermal necrolysis (TEN)
Not known: Subacute cutaneous lupus erythematosus

Musculoskeletal, connective tissue and bone disorders
Rare: Arthralgia, myalgia
Very rare: Muscular weakness
Uncommon: Fracture of the hip, wrist or spine

Renal and urinary disorders
Very rare: Interstitial nephritis, in some patients renal failure has been reported concomitantly

Reproductive system and breast disorders
Very rare: Gynaecomastia

General disorders and administration site conditions
Rare: Malaise, increased sweating

Other reported adverse events

Body as a Whole: abdomen enlarged, back pain, chest pain, substernal chest pain, facial edema, hot flushes, fatigue, flu-like disorder, generalized edema, leg edema, pain, rigors, allergic reaction, asthenia;

Cardiovascular: flushing, hypertension, tachycardia;

Endocrine: goiter;

Gastrointestinal: bowel irregularity, dyspepsia, dysphagia, dysplasia GI, epigastric pain, eructation, esophageal disorder, frequent stools, gastroenteritis, GI hemorrhage, GI symptoms not otherwise specified, hiccup, melena, mouth disorder, pharynx disorder, rectal disorder, serum gastrin increased, tongue disorder, constipation aggravated, ulcerative stomatitis, pancreatitis;

Hearing: earache, tinnitus;

Haematologic: anaemia, anaemia hypochromic, cervical lymphadenopathy, epistaxis, leukocytosis;

Hepatic: bilirubinemia;

Infections and Infestations: GI candidiasis, *Clostridium difficile* associated diarrhea

Metabolic/Nutritional: glycosuria, hyperuricemia, increased alkaline phosphatase, thirst, vitamin B_{12} deficiency, weight increase, weight decrease;

Musculoskeletal: arthritis aggravated, arthropathy, cramps, fibromyalgia syndrome, hernia, polymyalgia rheumatica, bone fracture;

Nervous System/Psychiatric: anorexia, apathy, appetite increased, hypertonia, nervousness, hypoesthesia, impotence, migraine, migraine aggravated, sleep disorder, tremor, depression aggravated, visual field defect;

Reproductive: dysmenorrhea, menstrual disorder, vaginitis;

Respiratory: asthma aggravated, coughing, dyspnea, larynx edema, pharyngitis, rhinitis, sinusitis;

Skin and Appendages: acne, pruritus ani, rash erythematous, rash maculo-papular, skin inflammation;

Special Senses: otitis media, parosmia;

Urogenital: abnormal urine, albuminuria, cystitis, dysuria, fungal infection, haematuria, micturition frequency, moniliasis, genital moniliasis, polyuria;

Visual: conjunctivitis;

Irritability, regurgitation and tachypnea have also been reported in paediatric patients.
Other clinically significant laboratory changes include:
Increased creatinine, hemoglobin, white blood cell count, platelets, potassium, sodium, thyroxine and thyroid stimulating hormone and decreased potassium and thyroxine

Endoscopic findings that were reported as adverse reactions include: duodenitis, esophagitis, esophageal stricture, esophageal ulceration, esophageal varices, gastric ulcer, gastritis, hernia, benign polyps or nodules, Barrett's esophagus, and mucosal discoloration.

Decreases in hemoglobin, white blood cell count and platelets.

USE IN SPECIAL POPULATIONS

- Pregnancy

For esomeprazole, clinical data on exposed pregnancies are insufficient. With the racemic mixture omeprazole data on a larger number of exposed pregnancies stemmed from epidemiological studies indicate no malformative nor foetotoxic effects. Reported animal studies with esomeprazole do not indicate direct or indirect harmful effects with respect to embryonal/fetal development. Reported animal studies with the racemic mixture do not indicate direct or indirect harmful effects with respect to pregnancy, parturition or postnatal development. Caution should be exercised when prescribing to pregnant women.

- Lactation

It is not known whether esomeprazole is excreted in human breast milk. No studies in lactating women have been reported. Therefore esomeprazole should not be used during breast-feeding.

- Paediatric

Esomeprazole should not be used in children younger than 12 years.

- Geriatrics

Dose adjustment is not required in the elderly

OVERDOSE

There is very limited experience to date with deliberate overdose. The symptoms described in connection with 280 mg were gastrointestinal symptoms and weakness. Single doses of 80 mg esomeprazole were uneventful. No specific antidote is known. Esomeprazole is extensively plasma protein bound and is therefore not readily dialyzable. As in any case of overdose, treatment should be symptomatic and general supportive measures should be utilised.

STORAGE:

Store below 30^{0}C, protected from moisture.

KEEP ALL MEDICINES OUT OF REACH OF CHILDREN. PACKING:

Packing

Cold form blister pack of 3x10's

Date of Revision: *February-2020*

Manufactured in India by:
Sun Pharmaceutical Industries Limited,
Paonta Sahib, Dist Sirmour
Himachal Pradesh – 173025

Product Registration Holder:
RANBAXY (MALAYSIA) SDN, BHD
(A Sun Pharma Company)
Lot 23, Bakar Arang Industrial Estate,
08000 Sungai Petani, Kedah, MALAYSIA